

The 2-Methylcitrate Cycle: A Reusable Propionate-Assimilation Module

A review-style synthesis for a molecular-biology audience Iteration 1 — commissioned review

1. Executive summary

The 2-methylcitrate cycle (2-MCC, also "methylcitrate cycle," MCC) is a compact metabolic module that oxidizes propionate to pyruvate and succinate, thereby both **assimilating** propionate as a carbon/energy source and **detoxifying** propionyl-CoA, a broadly inhibitory acyl-CoA. It is the dominant aerobic route of propionate catabolism in many bacteria and in fungi, and it operates as a citric-acid-cycle "mimic": propionyl-CoA (rather than acetyl-CoA) is condensed with oxaloacetate, and the resulting C7 tricarboxylate is processed by a series of reactions structurally parallel to citrate → *cis*-aconitate → isocitrate, but terminating in a carbon-carbon lyase step that regenerates a C4 (succinate) and releases a C3 (pyruvate)

([P 10482501](#)

12473114).

Five catalytic activities define the module: (1) **propionate activation** to propionyl-CoA by a PrpE-family CoA ligase; (2) **condensation** of propionyl-CoA with oxaloacetate to 2-methylcitrate by the citrate-synthase-family enzyme PrpC; (3) **dehydration** of 2-methylcitrate to 2-methyl-*cis*-aconitate — implemented either by the standalone, cofactorless dehydratase **PrpD**, or by the aconitase-like **AcnD** working with the accessory isomerase **PrpF**; (4) **hydration** of 2-methyl-*cis*-aconitate to 2-methylisocitrate by a **housekeeping aconitase** (AcnB or AcnA), not a dedicated enzyme; and (5) **cleavage** of 2-methylisocitrate to succinate + pyruvate by the lyase **PrpB**

([P 10482501](#)

11294638; 12473114; 29145506; 14575713).

The module's most important internal variation is at the dehydration step, where a stereochemical mismatch between PrpD chemistry and aconitase chemistry dictates whether an accessory isomerase (PrpF) is obligatory. Its most important external feature is that it plugs

into pre-existing central metabolism at two points — it borrows the general aconitase for hydration and returns products directly to the TCA cycle and gluconeogenesis. Because propionyl-CoA arises during infection (from host lipids, amino acids, cholesterol, and gut microbiota), the cycle is a virulence determinant in several pathogens and an attractive antimicrobial/antifungal target

([P 17973657](#));

31504787). In humans there is **no** 2-MCC; the appearance of methylcitrate in human biofluids is a pathological byproduct of propionic/methylmalonic acidemia, made by citrate synthase, and must not be conflated with the microbial cycle

([P 31451751](#)).

2. Definition and biological boundaries

2.1 What is inside the module

The 2-MCC is defined by the transformation **propionate + oxaloacetate → pyruvate + succinate**, carried out through a fixed set of intermediates:

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propionate →PrpE→ propionyl-CoA
propionyl-CoA + oxaloacetate →PrpC→ (2S,3S)-2-methylcitrate
2-methylcitrate →PrpD OR AcnD+PrpF→ 2-methyl-cis-aconitate
2-methyl-cis-aconitate →aconitase (AcnB/AcnA)→ (2R,3S)-2-methylisocitrate
2-methylisocitrate →PrpB→ succinate + pyruvate
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Everything required to convert free propionate into two central-metabolic products (pyruvate + succinate) belongs to the module. Oxaloacetate is consumed and succinate is returned, so the module is catalytic in C4-dicarboxylate in the same sense that the TCA cycle is catalytic in oxaloacetate: the net input is the propionyl unit, the net outputs are pyruvate + CO₂-equivalents once pyruvate is further oxidized.

2.2 What is adjacent but should be treated separately

- **The methylmalonyl-CoA pathway.** The other major propionate/propionyl-CoA route carboxylates propionyl-CoA (biotin-dependent propionyl-CoA carboxylase) to (S)-methylmalonyl-CoA and isomerizes it (adenosylcobalamin/B12-dependent methylmalonyl-

CoA mutase) to succinyl-CoA. This is the route in mammalian mitochondria and in many bacteria; it is **B12-dependent and CO₂-fixing**, whereas the 2-MCC is **B12-independent and oxidative**. The two are alternative solutions to the same problem and should be kept distinct

(
P 42341992
)

- **The TCA cycle and glyoxylate shunt.** The 2-MCC shares chemistry and even enzymes with these. The hydration step is done by the *same* aconitase that acts in the TCA cycle, and in *Pseudomonas aeruginosa* the glyoxylate-shunt isocitrate lyase (AceA) moonlights with 2-methylisocitrate lyase activity, linking the two cycles by enzymatic redundancy

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P 40015638
)

These overlaps are real biology, but the glyoxylate shunt (acetyl-CoA anaplerosis) and TCA cycle (acetyl-CoA oxidation) are separate modules with separate purposes.

- **Human "methylcitrate" chemistry.** In humans, propionyl-CoA accumulating in propionic and methylmalonic acidemia is condensed with oxaloacetate by ordinary citrate synthase to form methylcitrate, which is excreted as a **biomarker**, not processed by a cycle

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P 31451751
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16406646). There is no PrpD/PrpF/PrpB machinery in humans.

- **Regulatory and transport events.** The 2-methylcitrate *signal* (sensed by PrpR) and propionyl-CoA sensing (by the [4Fe4S]/CoA-binding PrpR of mycobacteria), as well as substrate transporters sometimes encoded in *prp* operons, are functionally linked but conceptually separate from the catalytic core

(
P 15528672
);

31504787; 41114503).

2.3 Competing definitions

The literature uses "2-methylcitric acid cycle," "2-methylcitrate cycle," and "methylcitrate cycle" interchangeably. A subtler definitional split concerns the **dehydration step**: some treatments describe a single "2-methylcitrate dehydratase" activity, but mechanistically there are two non-homologous implementations (PrpD; AcnD+PrpF) whose products are different aconitate isomers

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P 29145506
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17567742). A rigorous definition therefore treats "generation of 2-methyl-*cis*-aconitate" as a *function* that can be filled by either implementation, and treats the aconitase hydration as a **shared** step rather than a pathway-dedicated one.

3. Mechanistic overview

3.1 Step 1 — Propionate activation (PrpE)

Free propionate is activated to propionyl-CoA by an ATP-dependent propionyl-CoA synthetase of the PrpE (acyl-CoA synthetase/AMP-forming) family. In *Salmonella*, PrpE is encoded within the *prpBCDE* operon, but the cell also possesses **redundant** propionyl-CoA-forming systems — acetyl-CoA synthetase (Acs) and the propionate kinase/phosphotransacetylase (PduW/Pta) route — with different affinities that ensure propionyl-CoA supply across a range of propionate concentrations

([P 12700259](#)).

PrpE activity is itself post-translationally controlled by reversible acetylation (the CobB sirtuin/acetyltransferase system in *Salmonella*), tying activation to the cell's acetylation/energy status

([P 12700259](#));

9851993). Activation is **obligatory**: all downstream chemistry operates on the CoA thioester.

3.2 Step 2 — 2-Methylcitrate synthesis (PrpC)

PrpC, a member of the citrate-synthase family, condenses the methylene carbon of propionyl-CoA with the carbonyl of oxaloacetate to give (2S,3S)-2-methylcitrate. PrpC enzymes are typically **bifunctional**, retaining citrate synthase activity, but kinetic analysis shows propionyl-CoA is the preferred substrate

([P 10482501](#)).

This step is **obligatory and committing**: it defines the C7 methyl-branched tricarboxylate that the rest of the module processes, and its product (2-methylcitrate) is the physiological inducer of the pathway (Section 4.6).

3.3 Step 3 — Dehydration to 2-methyl-cis-aconitate (PrpD, or AcnD + PrpF)

This is the mechanistically richest and most variable step, and it exists in two alternative implementations that converge on the same product, 2-methyl-*cis*-aconitate:

- **Direct PrpD route.** PrpD is a **cofactorless** dehydratase — no iron–sulfur cluster, no metal, no reducing agent — unrelated in sequence to aconitases (P 11294638).

It removes water from (2S,3S)-2-methylcitrate by an **unusual *syn* elimination** to give *cis*-2-methylaconitate ($K_m \approx 440 \mu\text{M}$), and can also weakly hydrate *cis*-aconitate (~5-fold slower)

(P 12473114).

Crucially, PrpD does **not** hydrate 2-methyl-*cis*-aconitate to 2-methylisocitrate — that job belongs to the aconitase

(P 11294638).

PrpD belongs to the "MmgE/PrpD" family.

- **Two-component AcnD/PrpF route.** Many bacteria (e.g., *Shewanella oneidensis*) lack *prpD* and instead use **AcnD**, an aconitase-like (Fe-S) 2-methylcitrate dehydratase, together with **PrpF**. AcnD dehydrates 2-methylcitrate but yields a **different** 2-methylaconitate isomer (evidence points to 4-methylaconitate) that is **not** a substrate for the housekeeping aconitase. **PrpF**, a non-PLP isomerase structurally similar to diaminopimelate epimerase/proline racemase (with a catalytic Lys73 and Cys107), isomerizes the AcnD product into 2-methyl-*cis*-aconitate — the same isomer PrpD makes and the one aconitase accepts (P 17567742);

29145506). AcnD+PrpF fully complement a *Salmonella prpD* mutant, and PrpF is **obligatory** in this route (PrpD-using bacteria do not need PrpF)

(P 29145506).

The deep reason two implementations exist is **stereochemistry**: PrpD performs a *syn* elimination while aconitase adds water *anti*; the AcnD reaction produces the "wrong" geometric isomer, which is why an accessory isomerase is required to feed the shared aconitase step

(P 12473114);

29145506).

3.4 Step 4 — Hydration to 2-methylisocitrate (housekeeping aconitase, AcnB/AcnA)

The 2-methyl-*cis*-aconitate is hydrated to (2R,3S)-2-methylisocitrate by a **general-purpose aconitase**, not a pathway-dedicated enzyme. In *E. coli*, AcnB was purified from propionate-grown cells as the physiological 2-methylaconitate hydratase ($K_m \approx 210 \mu\text{M}$ for 2-methylisocitrate; no activity on 2-methylcitrate) (P 12473114).

In *Salmonella*, AcnA hydrates 2-methyl-*cis*-aconitate in vitro, yet *acnA* deletion does not block growth on propionate because AcnB provides redundant activity (P 11294638).

Thus the module **borrow**s an existing TCA-cycle activity for this step — an important boundary point, because the aconitase is shared with, and regulated by, central carbon/iron metabolism.

3.5 Step 5 — Carbon-carbon cleavage (PrpB)

PrpB, a 2-methylisocitrate lyase, cleaves (2R,3S)-2-methylisocitrate into **succinate + pyruvate** in a Mg^{2+} -dependent aldol-retro-cleavage reaction. Its crystal structures (from *E. coli* and *Salmonella*) place it in the isocitrate-lyase/PEP-mutase superfamily; the active site binds pyruvate and Mg^{2+} , with a mobile "active-site loop" that closes over the substrate, and subtle pocket differences rationalize why PrpB prefers 2-methylisocitrate while the related isocitrate lyase prefers isocitrate (P 14575713).

This step is **obligatory and defines the outputs**; succinate re-enters the TCA cycle and pyruvate enters central metabolism/gluconeogenesis.

3.6 Obligatory vs. conditional vs. accessory

- **Obligatory (core):** PrpE-type activation (or an equivalent propionyl-CoA source), PrpC condensation, a dehydration implementation, aconitase hydration, PrpB cleavage.
- **Conditional (implementation-dependent):** PrpF is required **only** in the AcnD-dependent route; PrpD and AcnD are mutually exclusive alternatives.

- **Accessory / context-dependent:** the choice of propionyl-CoA source (PrpE vs. Acs vs. PduW-Pta), CoA-transferase-based detoxification bypasses (fungi), transporters co-encoded in *prp* operons, and the regulatory proteins (PrpR).
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4. Major molecular players and active assemblies

Step	Enzyme / player	Family / fold	Cofactor	Notes
Activation	PrpE propionyl-CoA synthetase	AMP-forming acyl-CoA synthetase	ATP, Mg ²⁺	Redundant with Acs, PduW/Pta; regulated by acetylation (CobB) (P 12700259)
Condensation	PrpC 2-methylcitrate synthase	Citrate-synthase family	none	Bifunctional (citrate + methylcitrate synthase); prefers propionyl-CoA (P 10482501)
Dehydration (route A)	PrpD 2-methylcitrate dehydratase	MmgE/PrpD family	none (cofactorless)	syn elimination; no Fe-S; does not hydrate the product (P 11294638); 12473114)
Dehydration (route B)	AcnD + PrpF	AcnD: aconitase-like (Fe-S); PrpF: DAP-epimerase/proline-racemase fold	AcnD: [Fe-S]; PrpF: none	AcnD makes wrong isomer; PrpF (Lys73, Cys107) isomerizes to 2-methyl-cis-aconitate (P 17567742); 29145506)
Hydration	AcnB / AcnA aconitase	Aconitase (Fe-S)	[4Fe-4S]	Housekeeping TCA enzyme; redundant AcnA/AcnB (P 12473114); 11294638)
Cleavage	PrpB 2-methylisocitrate lyase	Isocitrate-lyase/PEP-mutase superfamily	Mg ²⁺	Products succinate + pyruvate; mobile active-site loop (P 14575713)

4.6 Regulatory assemblies (adjacent but tightly coupled)

- **PrpR (σ_{54} /bEBP type, γ -proteobacteria).** A σ_{54} -dependent enhancer-binding activator that senses **2-methylcitrate** (not propionate or citrate) as coactivator; it requires RpoN (σ_{54} /NtrA) and IHF to activate the *prpBCDE* promoter

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P 15528672
 10648513). Because PrpC must make 2-methylcitrate before the operon is induced, a small basal propionyl-CoA pool (from Acs/PduW-Pta) is needed to prime the system — a feed-forward logic

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P 12700259
 9851993).

- **PrpR (*Mycobacterium tuberculosis*).** A structurally distinct transcription factor that binds CoA/short-chain acyl-CoAs and carries a **[4Fe4S] cluster**; it activates *prpC/prpD* during cholesterol catabolism. The cluster is stable (probably an iron/nutrient sensor rather than a redox sensor)

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P 31504787
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5. Evolutionary and cell-biological variation

5.1 Across bacterial lineages

- **γ -Proteobacteria (*E. coli*, *Salmonella*).** Canonical *prpBCDE* operon; PrpD route; hydration by AcnB (with redundant AcnA); σ_{54} /PrpR/IHF regulation

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P 10482501
 10648513; 12473114).

- **AcnD/PrpF users (e.g., *Shewanella*).** Replace *prpD* with *acnD* + *prpF*; obligate isomerase requirement

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P 29145506
).

- **Actinobacteria (*Corynebacterium glutamicum*).** Two paralogous *prpDBC* clusters (*prpD1B1C1*, *prpD2B2C2*); the *prpD2B2C2* cluster is the propionate-induced, physiologically

relevant one; PrpC1/PrpC2 are both active citrate/methylcitrate synthases

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P 11976302
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- **Mycobacterium tuberculosis.** MCC is up-regulated to detoxify propionyl-CoA generated by **cholesterol** catabolism during infection; controlled by the [4Fe4S]/acyl-CoA-sensing PrpR — a key host-adaptation circuit

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P 31504787
).

- **Pseudomonas aeruginosa.** MCC is functionally interwoven with the glyoxylate shunt; isocitrate lyase (AceA) supplies secondary 2-methylisocitrate lyase activity, an example of cross-pathway enzymatic redundancy

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P 40015638
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- **Acetic-acid bacteria (Acetobacter).** *prp* clusters frequently **lack a dedicated prpE**, recruiting a PrpE-homologous acetyl-CoA synthetase (*prpE'*) into the operon; growth on propionate requires PrpB, PrpC and PrpD

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P 42271202
).

- **Neisseria meningitidis.** The *prp* operon includes a transporter for α -ketobutyrate, and the pathway balances propionate **assimilation vs. detoxification** depending on the host microenvironment and co-substrate

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P 41114503
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5.2 Fungi

Filamentous fungi run the MCC to detoxify propionyl-CoA arising from amino acids (Val/Ile/Met), odd-chain fatty acids, and propionate. Genetic proof exists for *Aspergillus nidulans* and *A. fumigatus*, and for pathogenic *Fusarium* species

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P 19661181
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16008561). Fungal methylcitrate synthase (**mcsA**) is the pivotal enzyme: its deletion causes propionyl-CoA accumulation, blocks polyketide synthesis and conidial pigmentation, and directly inhibits the **pyruvate dehydrogenase complex**

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P 16008561
).

Fungi possess an additional **CoA-transferase** detoxification bypass that transfers CoASH from propionyl-CoA to acetate, relieving toxicity independently of the cycle

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P 18331473
).

MCC is a **virulence factor**: *A. fumigatus mcsA* mutants are attenuated in murine invasive aspergillosis and cleared by the host, marking the cycle as an antifungal target

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P 17973657
).

MCC activation also remodels the cell wall and increases oxidative stress in *Histoplasma capsulatum*

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P 40023528
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5.3 Compartmentalization

In fungi (and by analogy in eukaryotic MCC-bearing lineages) propionyl-CoA metabolism and the MCC are associated with the **mitochondrion**, co-localized with the source of oxaloacetate and the TCA machinery. In bacteria the module is cytoplasmic. The compartment dictates substrate access (oxaloacetate, CoA pools) and coupling to respiration.

5.4 Humans and mammals — a boundary, not a variant

Mammals do **not** operate the 2-MCC; they use the B12-dependent methylmalonyl-CoA route. When that route fails (propionic/methylmalonic acidemia), propionyl-CoA accumulates and citrate synthase produces **methylcitrate**, a diagnostic biomarker and a toxic metabolite that inhibits TCA and urea-cycle enzymes — but the downstream PrpD/aconitase/PrpB steps do not exist in humans

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P 31451751
);

16406646; 32451238).

6. Constraints, dependencies, and failure modes

6.1 Ordering constraints

The chemistry enforces a strict order: activation → condensation → dehydration → hydration → cleavage. Each enzyme's substrate is the previous enzyme's product, and the intermediates are not interconvertible by other routes. The **dehydration–hydration pair** is a formal dehydration/rehydration (as in citrate → isocitrate via aconitate) that repositions the hydroxyl to make the C–C bond cleavable by PrpB.

6.2 Stereochemical/isomer constraints (mutual exclusivity)

- Aconitase hydrates only 2-methyl-*cis*-aconitate; it is inactive on the AcnD product. Therefore AcnD **cannot** function without PrpF

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P 29145506

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- PrpD and AcnD are **mutually exclusive** implementations of the same step; genomes carry one or the other, and PrpF co-occurs with AcnD, not with PrpD

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P 17567742

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- AcnB hydrates 2-methyl-*cis*-aconitate but shows no activity on 2-methylcitrate; substrate specificity is strict

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P 12473114

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6.3 Regulatory dependency (feed-forward)

The *prpBCDE* operon is induced by 2-methylcitrate, which can only be made by PrpC from propionyl-CoA. So a **basal propionyl-CoA supply** (Acs, PduW/Pta) is a prerequisite to switch on the full operon — a chicken-and-egg dependency the cell resolves with redundant activation systems

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P 12700259

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9851993). In mycobacteria the analogous switch is an acyl-CoA/[4Fe4S] sensor

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P 31504787

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6.4 Failure modes

- **Loss of the cycle → propionyl-CoA toxicity.** Blocking PrpC/mcsA causes propionyl-CoA accumulation that inhibits pyruvate dehydrogenase, polyketide synthesis, growth, and (in pathogens) virulence

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P 16008561
 17973657).

- **Metabolic imbalance from intermediate accumulation.** Accumulated 2-methylcitrate itself is inhibitory/imbalancing

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P 41684677
 and, in the human analogue, methylcitrate inhibits TCA/urea-cycle enzymes,

P 16406646).

- **Redundancy buffers single losses.** *acnA* loss is masked by AcnB; *prpB* loss can be partly covered by AceA in *P. aeruginosa*; missing *prpE* is covered by *prpE'*/Acs

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P 11294638
 40015638; 42271202). This redundancy is a robustness feature but complicates single-gene knockout interpretation.

7. Controversies and open questions

1. **Identity of the AcnD product.** The AcnD reaction yields a 2-methylaconitate isomer that is not an aconitase substrate; indirect evidence (in vivo/in vitro, PrpF requirement) points to **4-methylaconitate**, but the precise structure/stereochemistry has been inferred rather than directly determined for every system

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P 29145506
).

2. **Which aconitase does the hydration in vivo.** In *E. coli* it is AcnB; in *Salmonella* AcnA is competent but AcnB provides redundancy — the physiological split between AcnA and AcnB, and how iron/oxidative status shifts it, is not fully resolved

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P 12473114
 11294638).

3. **Boundaries between the MCC and the glyoxylate shunt.** The AceA/PrpB and AcnA/AcnB overlaps show the "cycle" is not cleanly separable from central metabolism; how flux is partitioned under infection-relevant conditions is an open, organism-specific question (P 40015638).
4. **Cross-organism extrapolation.** Much mechanistic detail comes from a few models (*Salmonella*, *E. coli*, *Shewanella*, *Aspergillus*). Whether kinetic and stereochemical conclusions transfer to actinobacteria, acetic-acid bacteria, and diverse fungi is often assumed rather than demonstrated (P 11976302); 42271202).
5. **Regulatory diversity.** The σ_{54} /PrpR (γ -proteobacteria) and [4Fe4S]/acyl-CoA PrpR (mycobacteria) are unrelated solutions to sensing propionate flux; the full range of MCC regulatory logic across lineages (and nitrogen/GlnR control) is incompletely mapped (P 15528672); 31504787; 30745367).
6. **Therapeutic targeting.** Because the MCC is essential for virulence in *A. fumigatus* and important in *M. tuberculosis* cholesterol catabolism yet absent in humans, its enzymes (PrpC/mcsA, PrpB, PrpD) are candidate drug targets; selective inhibitors and resistance via redundancy remain to be worked out (P 17973657); 31504787).

8. Key references

- Horswill AR, Escalante-Semerena JC. *Salmonella typhimurium* LT2 catabolizes propionate via the 2-methylcitric acid cycle.

(P 10482501)

(1999).

- Horswill AR, Escalante-Semerena JC. In vitro conversion of propionate to pyruvate: PrpD and aconitase.

P 11294638

(2001).

- Brock M et al. Oxidation of propionate to pyruvate in *E. coli*: methylcitrate dehydratase and aconitase (AcnB); *syn/anti* stereochemistry.

P 12473114

(2002).

- Garvey GS et al. Crystal structure of PrpF (*Shewanella*): a non-PLP isomerase.

P 17567742

(2007).

- Rocco CJ et al. PrpF isomerizes 2-methyl-*cis*-aconitate in the AcnD-dependent 2-MCC; AcnD product likely 4-methylaconitate.

P 29145506

(2017).

- Simanshu DK et al. Crystal structure of *Salmonella* PrpB (2-methylisocitrate lyase) with pyruvate/Mg²⁺.

P 14575713

(2003).

- Palacios L, Escalante-Semerena JC. PrpR senses 2-methylcitrate.

P 15528672

(2004).

- Palacios L, Escalante-Semerena JC. prpR, ntrA (σ 54), ihf required for *prpBCDE* expression.

P 10648513

(2000).

- Palacios L et al. Propionyl-CoA is the common intermediate; redundant activation systems.

P 12700259

(2003).

- Tsang AW et al. Regulation integrating 1,2-propanediol and propionate catabolism.

P 9851993

(1998).

- Claes WA et al. Two *prpDBC* clusters in *Corynebacterium glutamicum*.

P 11976302

(2002).

- Maerker C et al. *A. fumigatus* methylcitrate synthase; propionyl-CoA inhibits PDH and polyketide synthesis.

P 16008561

(2005).

- Ibrahim-Granet O et al. Methylcitrate synthase essential for invasive aspergillosis.

P 17973657

(2008).

- Fleck CB, Brock M. CoA-transferase detoxification of propionyl-CoA in *A. nidulans*.

P 18331473

(2008).

- Domin N et al. Functional MCC in *Fusarium*.

P 19661181

(2009).

- Tang J et al. *M. tuberculosis* PrpR: [4Fe4S]/acyl-CoA-sensing transcription factor.

P 31504787

(2019).

- Wijaya F et al. MCC and glyoxylate cycle linked by enzymatic redundancy in *P. aeruginosa* (AceA secondary 2-MIC lyase).

P 40015638

(2025).

- Talà A et al. Methylcitrate pathway balancing assimilation/detoxification in meningococci; α -ketobutyrate transporter.

P 41114503

(2025).

- Yang B et al. *prp* operon in *Acetobacter pasteurianus*; *prpE'* recruitment.

P 42271202

(2026).

- Santos LPA et al. Propionate/MCC in *Histoplasma capsulatum*.

P 40023528

(2025).

- Al-Dirbashi OY et al. Methylcitrate/citrate ratio as biomarker of inborn propionate errors (human, non-cyclic).

P 31451751

(2019).

- Filipowicz HR et al. Metabolic changes in propionic acidemia (human methylcitrate).

P 16406646

(2006).

Uncertainty statement

Mechanistic conclusions on PrpD/AcnD/PrpF stereochemistry and aconitase involvement rest chiefly on *E. coli*, *Salmonella*, and *Shewanella*; fungal MCC on *Aspergillus* and *Fusarium*. The exact structure of the AcnD product, the in-vivo AcnA/AcnB partition, and the flux boundary with the glyoxylate shunt are inferred or organism-specific. Human "methylcitrate" is a pathological citrate-synthase product, **not** evidence of a 2-MCC in mammals. Claims here should not be uniformly extrapolated across all lineages.